APPLICATION: 3

HUFFMANN CODING

Huffman Coding

- It is one of the application of Binary tree.
- Huffman Coding is a famous **Greedy Algorithm**.
- It is used for the **lossless compression** of data.
- It uses **variable length** encoding.
- It assigns variable length codes to all the characters depending on how frequently they occur in the given text.
- The character which occurs **most frequently** gets the **smallest** code.
- The character which occurs **least frequently** gets the **largest** code.
- Message which is **repeating more** number of times is **near to ROOT**.
- It is also known as **Huffman Encoding**.

Huffman Coding

Major Steps in Huffman Coding-

There are two major steps in Huffman coding-

1. Building a Huffman tree from the input characters.
2. Assigning codes to the characters by traversing the Huffman tree.

Huffman Coding

The steps involved in the construction of Huffman Tree are as follows-

Step-01:

- Create a leaf node for all the given characters.
- Leaf node contains the occurring frequency of that character.

Step-02:

- Arrange all the nodes in increasing order of their frequency value.

Huffman Coding

Step-03:

Considering the first two nodes having minimum frequency,

- Create a new internal node.
- The frequency of this node is the sum of the frequency of those two nodes.
- Make the first node as a left child and the other node as a right child of the newly created node.

Step-04:

- Keep repeating Step-02 and Step-03 until all the nodes form a single tree.
- The tree finally obtained will be the desired Huffman Tree.

Huffman Coding

Important Formulas-

Formula-01:

$$\text{Average code length per character} = \frac{\Sigma\,(\text{frequency}_i \times \text{code length}_i)}{\Sigma\,\text{frequency}_i}$$

$$= \Sigma\,(\text{probability}_i \times \text{code length}_i)$$

Huffman Coding

Formula-02:

Total number of bits in Huffman encoded message

= Total number of characters in the message x Average code length per character

$= \sum (\text{frequency}_i \times \text{Code length}_i)$

Huffman Coding

PRACTICE PROBLEM BASED ON HUFFMAN CODING-

Problem-

A file contains the following characters with the frequencies as shown. If Huffman coding is used for data compression, determine-

1. Huffman code for each character
2. Average code length
3. Length of Huffman encoded message (in bits)

Huffman Coding Example

Characters	Frequencies
a	10
e	15
i	12
o	3
u	4
s	13
t	1

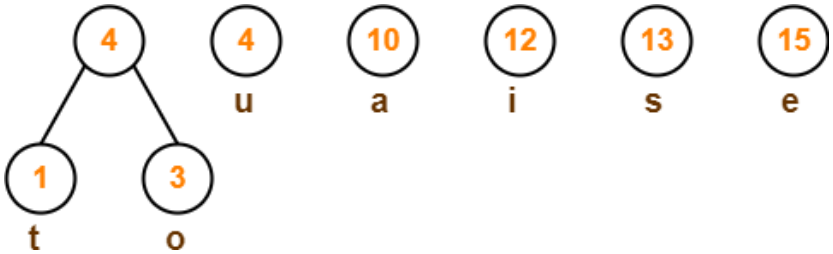
Characters	A	E	I	O	U	S	t
frequency	10	15	12	3	4	13	1

Solution-

First let us construct the Huffman Tree.
Huffman Tree is constructed in the following steps-

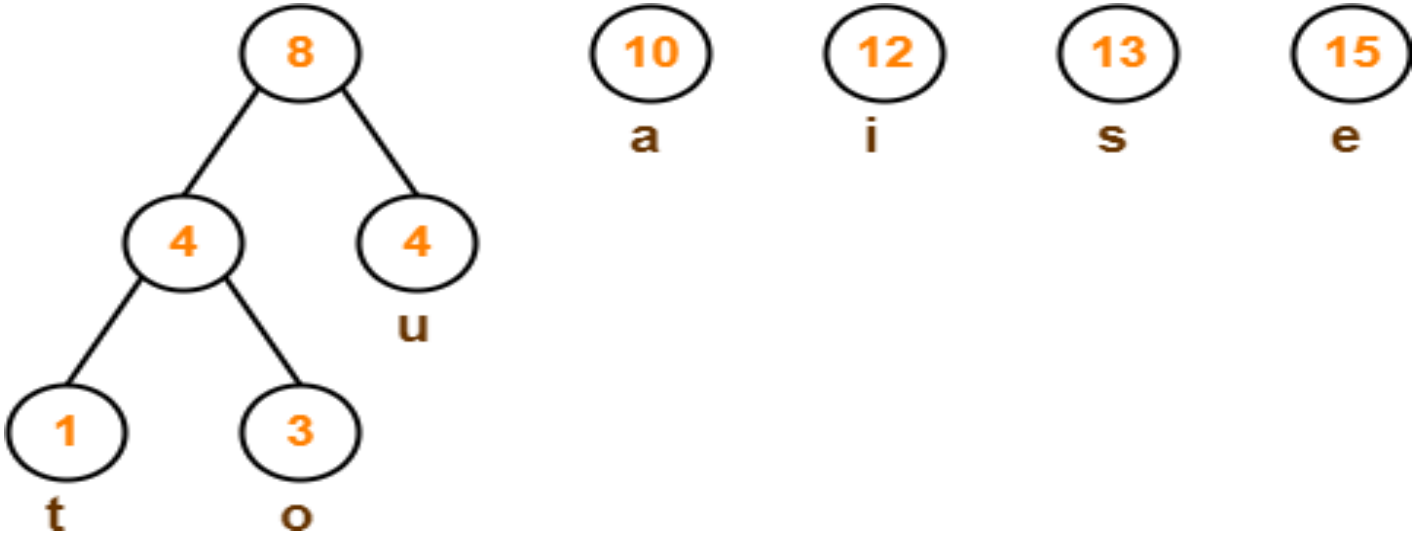
Step-01:

Step-02:



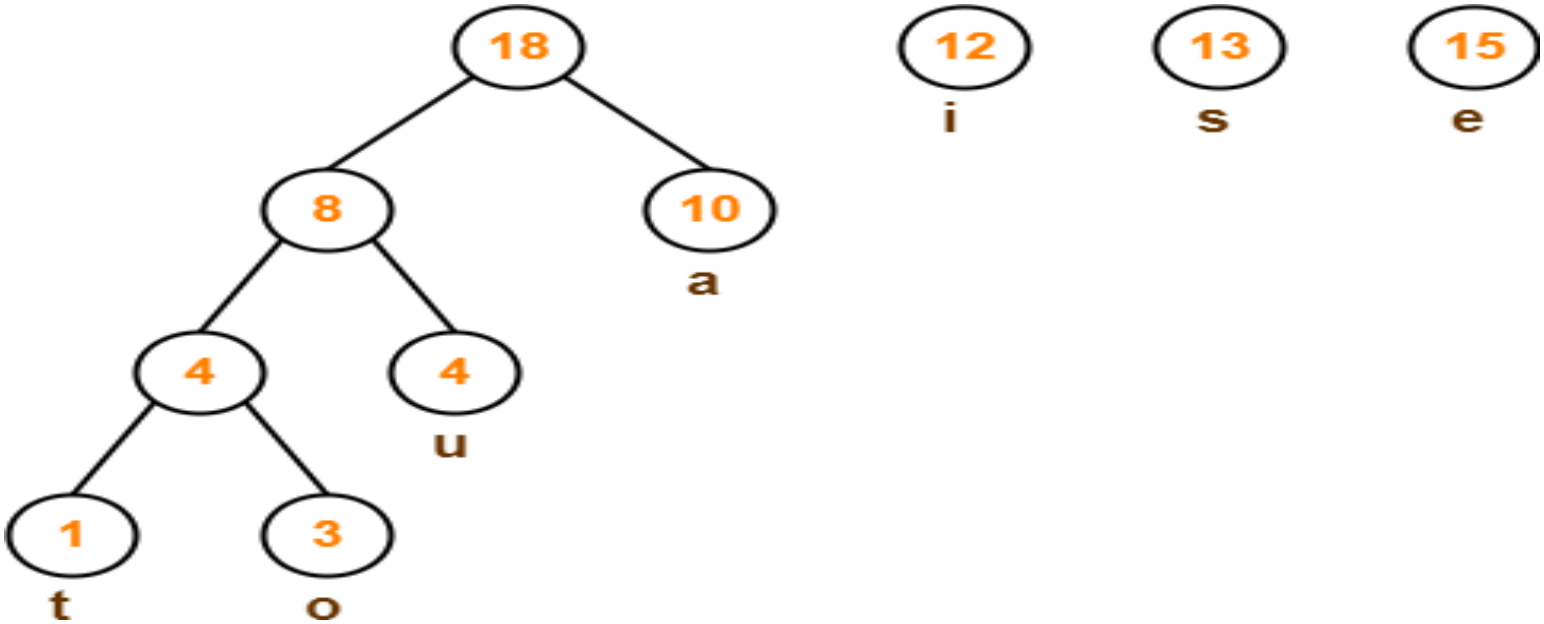
Huffman Coding

Step-03:



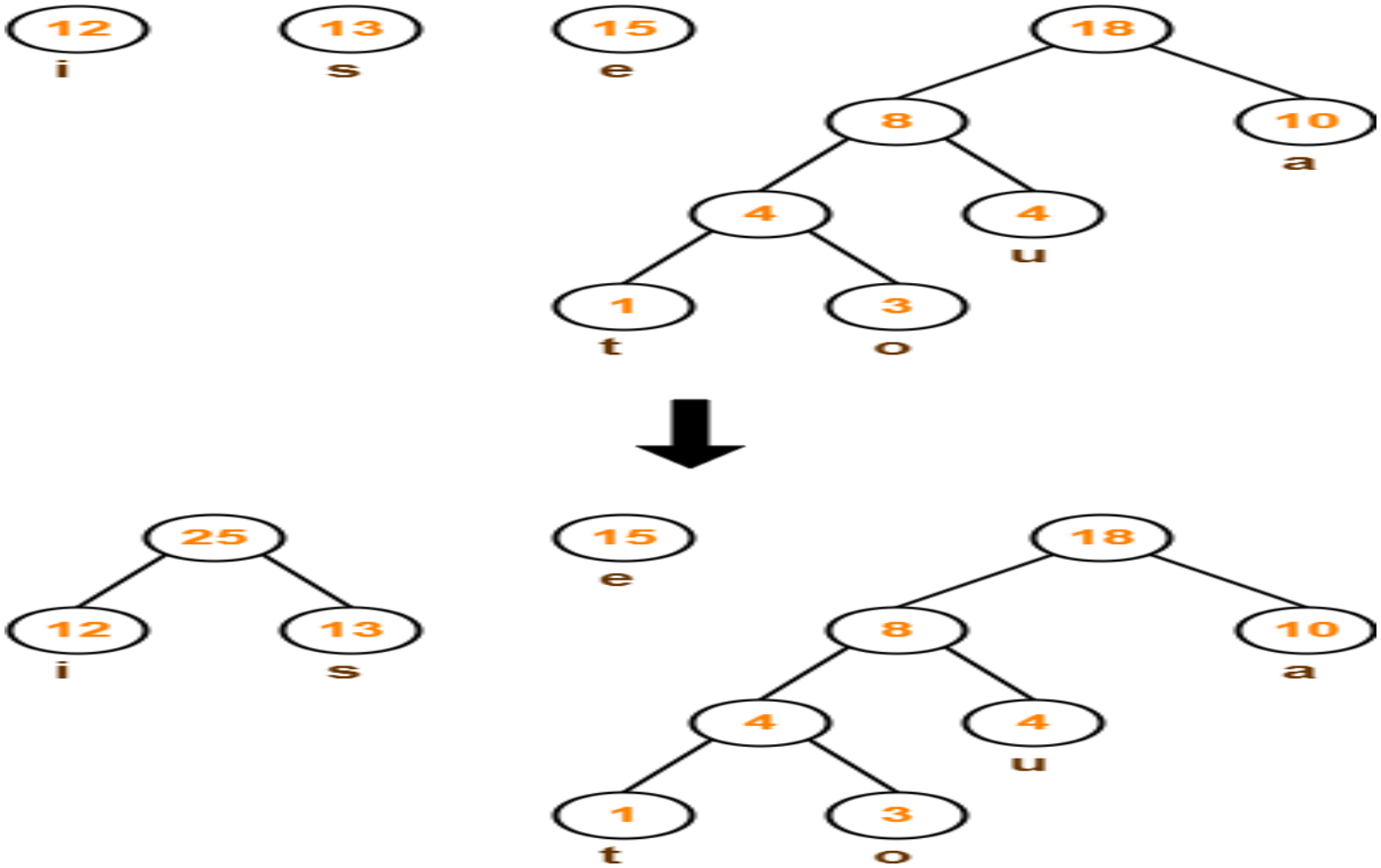
Huffman Coding

Step-04:



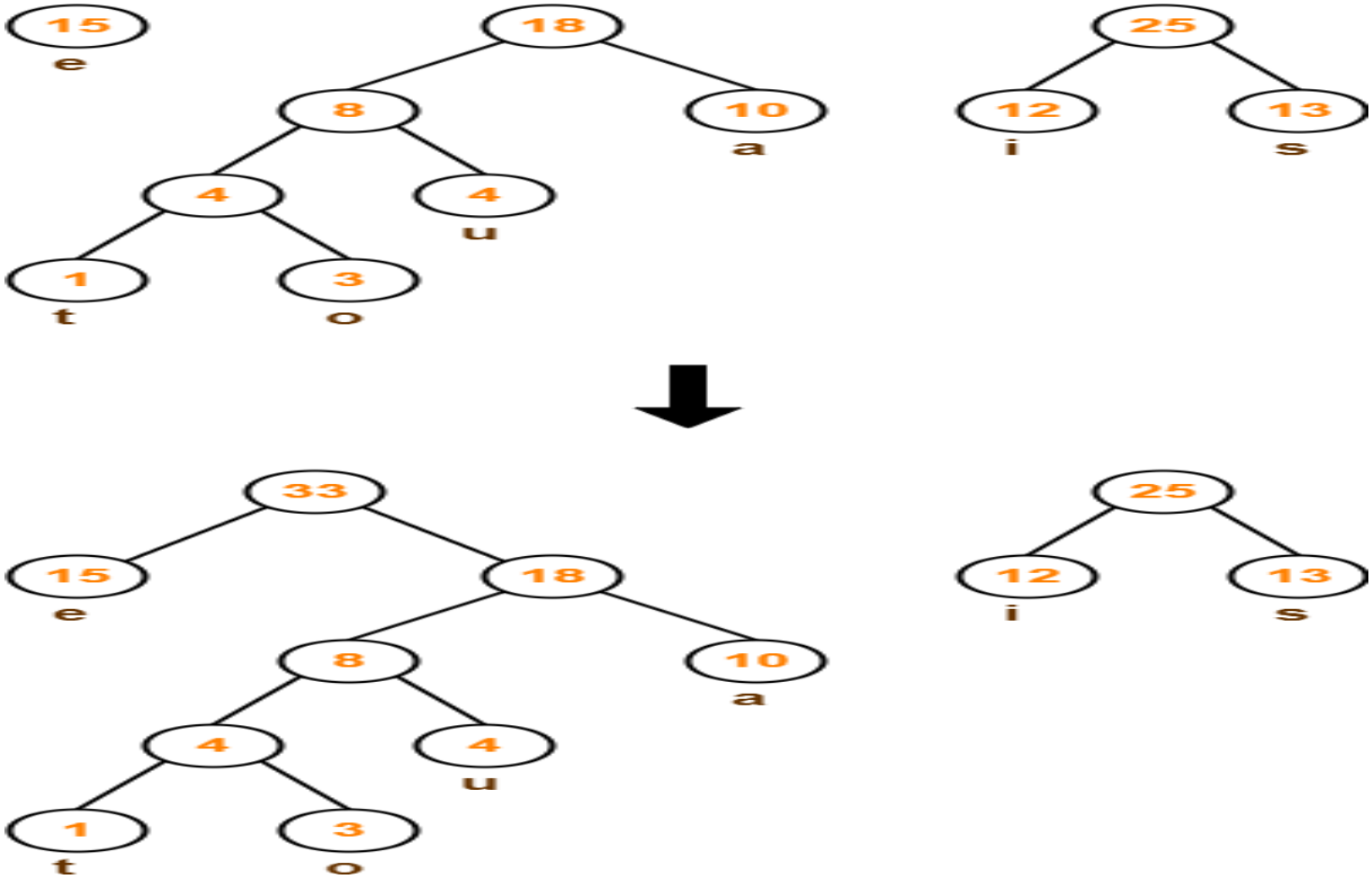
Huffman Coding

Step-05:



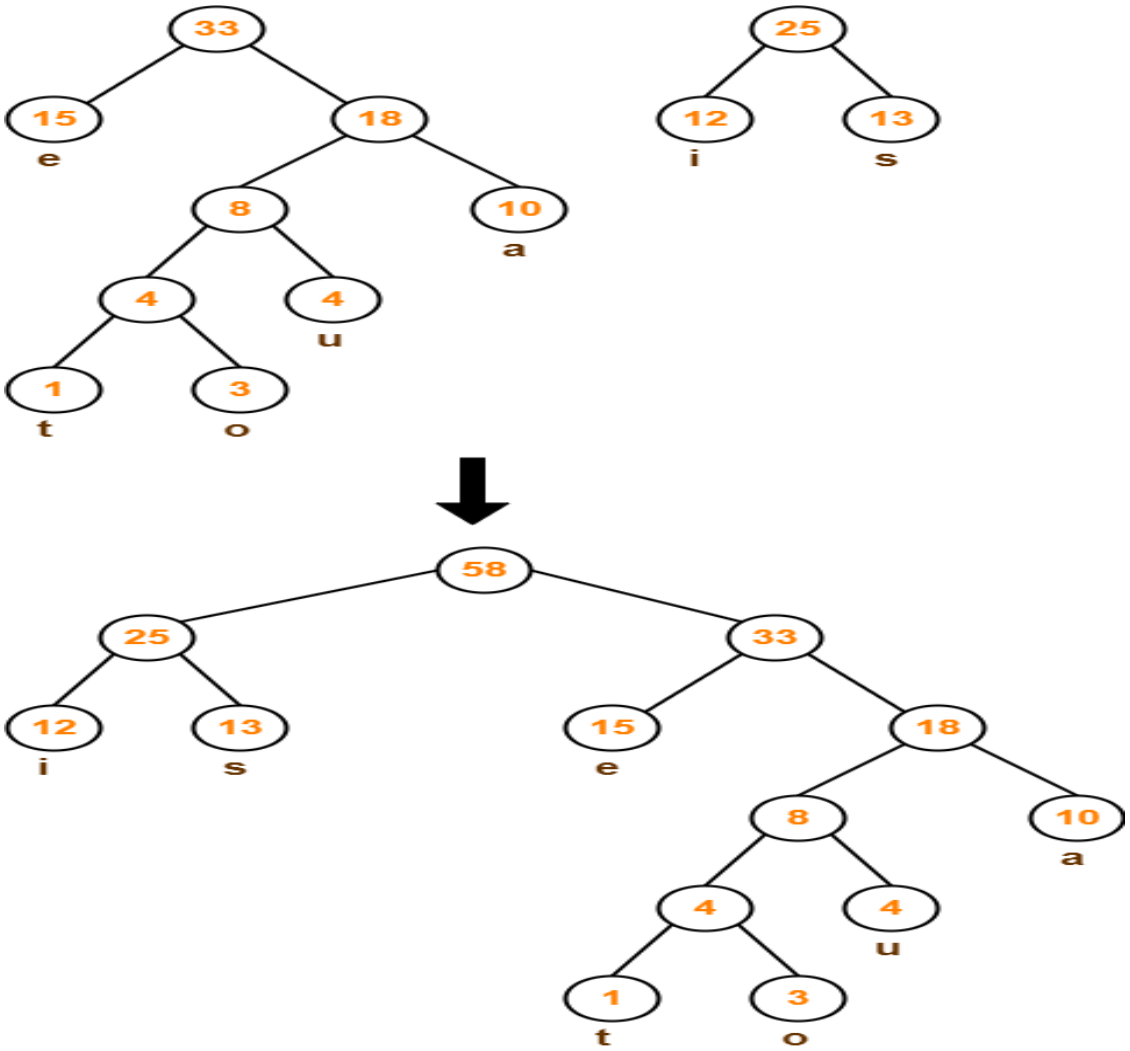
Huffman Coding

Step-06:



Huffman Coding

Step-07:



Huffman Tree

Huffman Coding

Now,

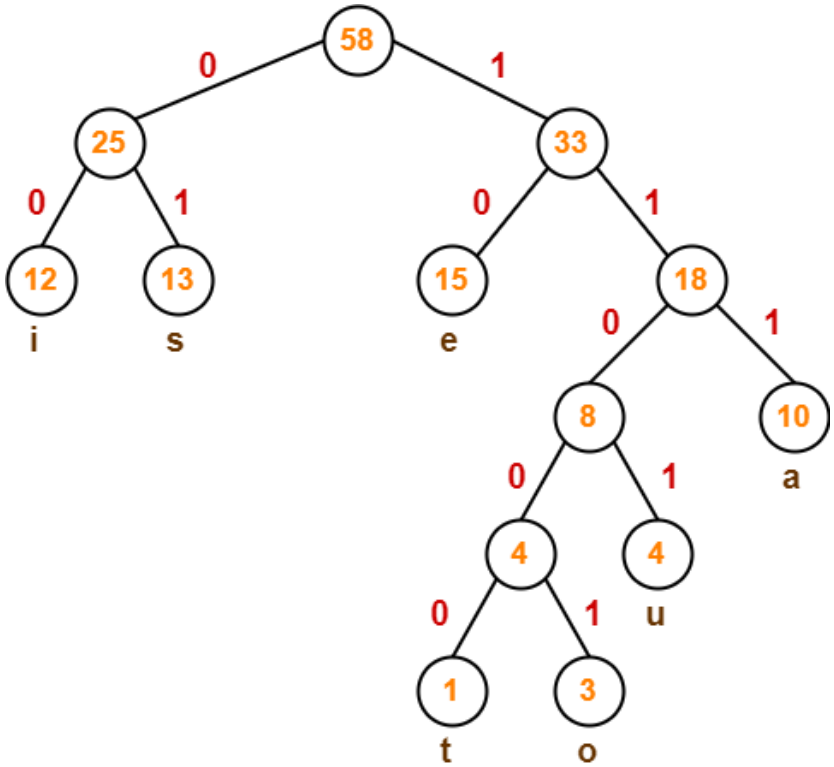
- We **assign weight** to all the edges of the constructed Huffman tree.
- Let us assign weight **'0' to the left** edges and weight '**1' to the right** edges.

Rule

- If you assign weight '0' to the left edges, then assign weight '1' to the right edges.
- If you assign weight '1' to the left edges, then assign weight '0' to the right edges.
- But follow the same convention at the time of decoding that was adopted at the time of encoding.

After assigning weight to all the edges, the modified Huffman Tree is-

Huffman Coding



Huffman Tree

Now, let us answer each part of the problem one by one-

Huffman Coding

1. Huffman Code For The Characters-

To write Huffman Code for a character, traverse the Huffman tree from root node to all the leaf node of that character.

Following this rule, **the Huffman Code for each character** is-

- a = 111
- e = 10
- i = 00
- o = 11001
- u = 1101
- s = 01
- t = 11000

Huffman Coding

From here, We can observe-

- Characters occurring less frequently in the text are assigned the larger code.
- Characters occurring more frequently in the text are assigned the smaller code.

2. <u>Average Code Length-</u>

Average code length

$= \sum (\text{frequency}_i \times \text{code length}_i) / \sum (\text{frequency}_i)$

$= \{ (10 \times 3) + (15 \times 2) + (12 \times 2) + (3 \times 5) + (4 \times 4) + (13 \times 2) + (1 \times 5) \} / (10 + 15 + 12 + 3 + 4 + 13 + 1)$

$= 2.52$

Huffman Coding

3. <u>Length of Huffman Encoded Message-</u>

Total number of bits in Huffman encoded message

= Total number of characters in the message x Average code length per character

= 58 x 2.52

= 146.16

≅147 bits